

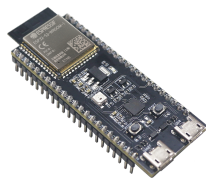
Microphone Array

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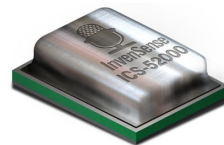
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Overview

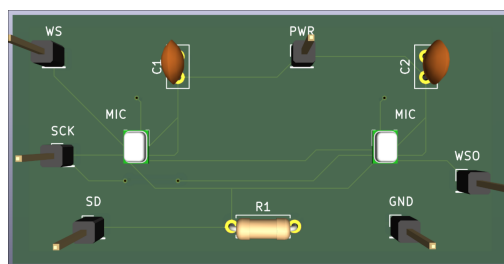
I'm building a microphone array that can locate the source of an audio signal with beamforming. The design consists of equally spaced ICS-52000 microphones soldered to PCBs, see figure 1. These are in turn connected to an ESP32-S3 DevKitC development board for signal processing. Using beamforming the development board will approximate the position of the audio source and output it to a laptop or phone.



(a) ESP32-S3 DevKitC



(b) ICS-52000



(c) PCB prototype

Figure 1: Components currently in use for the Microphone Array

Design

The system utilizes a rectangular array geometry to provide high angular resolution across both horizontal and vertical axes. By employing ICS-52000 digital microphones, the design leverages the TDM (Time Division Multiplexing) protocol, allowing multiple audio channels to be daisy-chained on a single bus.

This configuration minimizes the wiring complexity while maintaining high-fidelity 24-bit signal capture. The spatial separation between microphones is calculated to optimize the Beamforming algorithm, ensuring the phase difference between incoming signals is sufficient for accurate Time Difference of Arrival (TDOA) calculations without causing spatial aliasing at target frequencies.

Progress

As of the last update the development board and the microphones have been bought. The PCB shown in figure 1 (designed in KiCAD) has been ordered from JLCPCB. The first phase of development will consist of me setting up two microphones and writing the C++ code for them to be able to differentiate if sound is coming from the left or from the right.

Roadmap

The project is structured into four primary development phases to transition from hardware validation to intelligent signal processing:

- **Phase 1: Hardware Verification & Dual-Channel Logic**
Initial assembly of the custom PCBs and validation of the I2S/TDM bus. This phase focuses on writing the C++ drivers to synchronize two microphones and implementing basic phase-comparison logic to differentiate sound direction (left vs. right).
- **Phase 2: Full Array Scaling**
Expansion of the firmware to support the full rectangular grid. Implementation of the Delay-and-Sum beamforming algorithm on the ESP32-S3 to calculate 2D spatial coordinates of the primary sound source.
- **Phase 3: Real-Time Visualization**
Developing a communication bridge to output approximated coordinates to a laptop or mobile device. This includes creating a simple graphical interface to visualize the localized audio source in real-time.
- **Phase 4: Machine Learning Integration**
Leveraging the ESP32-S3's AI acceleration instructions to implement a classification model. The goal is to move beyond simple localization and begin identifying the specific nature of the audio source (e.g., voice recognition or anomaly detection).